Bhuvan Arora

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RESEARCH INTERESTS

Evaluating and training LLM agents on long-horizon tasks, reinforcement-learning environments built from real-world settings, multilingual and Indic NLP, and cultural bias in large language models.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani 2023 – May 2027 (expected)
B.E. Computer Science · **CGPA 9.57 / 10**
Relevant coursework: Machine Learning, Introduction to LLMs, Design and Analysis of Algorithms, Probability and Statistics, Theory of Computation

PUBLICATIONS AND PREPRINTS

Bhuvan Arora, Chaitanya Jain, Dilesh Churiwal, Ankita Karmakar, Manish Prasad, Raja Piyush, Harshit Jain, Shivansh Dubey, Dhruv Kumar, Raunak Manekar. **KiranaBench: Can a Language Model Run a Neighbourhood Shop?** Under review at ACL Rolling Review (October 2026 cycle), committing to NAACL 2027.

Bhuvan Arora, Devesh Saraogi, Sravya Varada, Dhruv Kumar. **DiSCo: A Distribution-First Steering and Cultural Prior Evaluation Framework for Measuring Cultural Preference Bias in LLMs.** arXiv:2609.10253, 2026. Received a major-revision decision at a peer-reviewed venue.

Siva Sai, **Bhuvan Arora**, Vineet Suri, Vinay Chamola, Mohan Gurusamy. **LLM-Driven Security and Resilience in 6G Mission-Critical Communication Networks.** *IEEE Open Journal of the Communications Society*, vol. 7, pp. 8963–8984, 2026 (survey). doi:10.1109/OJCOMS.2026.3717648

RESEARCH EXPERIENCE

Karobaar: a suite of Indian business simulations as RL environments for LLM agents 2026 – present
Advisor: Dr. Dhruv Kumar, BITS Pilani · Project lead

- Leading a suite of simulated Indian businesses in which LLM agents run the business over a long horizon, scored against simulator ground truth: kirana shop, truck fleet, municipal corporation, wedding planning, staffing agency, scrap dealer, hotel and cloud kitchen.
- Built **bizgym**, the shared core: one plug-in per business, with the simulator's own score as the reward (no LLM judge). One-day episodes from saved situations, compared in groups (GRPO-style advantages).
- Training Qwen3.5-4B with RL on the kirana environment; behaviour-cloning and RL runs in progress.

BharatGUI: Indic-script GUI environments for computer-use agents 2026 – present
Advisor: Dr. Dhruv Kumar, BITS Pilani

- Building interactive, state-verified environments to measure how screenshot-based computer-use agents degrade when an interface is fully rendered in an Indic script, including reading native-script values and typing them into Latin-only fields.
- First environment built: a government-service portal sandbox (form, upload, review, reference number, status tracking).

KiranaBench: can a language model run a neighbourhood shop? 2026
Advisor: Dr. Dhruv Kumar, BITS Pilani · First author

- Built a simulated Indian general store in which one model holds every job (manager, cashier, helper and munshi, the bookkeeper) for 30 simulated days. It learns the day only from what its own staff report.
- Customers and suppliers, voiced by a fixed environment model, deal with the shop in English, Hindi or Hinglish; demand is drawn from India's national household consumption survey; every decision is scored against deterministic rules.
- Four open-weight models fall well short: the best earns an eighth of what a fixed-rule shopkeeper does, two finish below a shop nobody manages, and every model writes worse to Hindi and Hinglish suppliers than to English ones.

DiSCo: measuring cultural preference bias in LLMs 2026
Advisor: Dr. Dhruv Kumar, BITS Pilani · First author

- Developed a distribution-first evaluation framework for how LLMs handle culturally specific norms and values, beyond language differences alone.

Multilingual robustness of agentic tool-calling in Indic languages Ongoing

Advisor: Dr. Dhruv Kumar, BITS Pilani

- Studying how reliably LLM agents use tools (search, APIs, scheduling) when instructions and context are in Indic languages rather than English. Currently constructing the dataset.

Physical AI companion for older adults Ongoing

Advisor: Dr. Siddharth Mehrotra

- Built a working prototype of a voice assistant operated with physical buttons and knobs, designed for older adults with limited digital literacy. It supports open-ended conversation, web search, and reminders and alarms.
- Building an internet-based calling, voice-note and messaging backend that connects users to family members through a companion app.

INDUSTRY EXPERIENCE

Samsung R&D Institute India, Bangalore (SRI-B): Advanced Developer Intern May – Jul 2026

MAGPIE summer internship · Received a pre-placement offer

- **AI planner app:** conversational daily planner using Google's A2UI (Agent-to-UI) protocol for input and confirmation, with location-aware clash detection (Nominatim, OpenRouteService). Tested whether A2UI, designed for remote agents, can run on-device; it partially worked.
- **Damaged-barcode recovery:** two-stage retrieval pipeline (FFT-based embeddings with approximate nearest-neighbour shortlisting, then cross-correlation reranking) that reached **95.75% accuracy across 9 barcode types**.

CloudDefense.AI: Practice School I intern May – Jul 2025

OTHER PROJECTS

- **Traffic-signal RL under sensor failures** (2026): how missing, noisy and delayed sensor readings affect RL traffic-signal controllers (SUMO, PPO/DQN), and which remedies help.
- **local-flow** (2026): open-source dictation tool that runs fully locally. GitHub
- **AI Buddies Multi-Agent System** (2025): multi-agent LLM application. GitHub
- **BITS Connect** (2025): Flutter and Firebase app that lets BITS Pilani students form study groups and meetups. GitHub

SKILLS AND CERTIFICATIONS

- **Research:** LLM agents and tool-calling; simulation-based benchmarks and RL environments; RL fine-tuning (GRPO-style, behaviour cloning); multilingual evaluation
- **Tools:** Python; vLLM, Hugging Face Transformers, PEFT, Ollama; ElevenLabs voice stack; Flutter, Firebase
- **Certifications:** NVIDIA DLI Fundamentals of Deep Learning (2025)